

PROJECT TITLE

RETAILX ANALYTICS SUITE

DOMAIN

BUSINESS INTELLIGENT AND DATA ANALYTICS

TOOLS USED

MICROSOFT EXCEL

MICROSOFT POWER BI

DAX

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Project Overview

RetailX Analytics Suite is a comprehensive business intelligence solution designed to analyze and visualize sales performance, customer behavior, and inventory operations. The project integrates Excel as a data source and Power BI for advanced data modeling and interactive dashboard development.

The system enables stakeholders to monitor key business KPIs such as revenue, profit, customer distribution, and inventory levels in real time, helping improve decision-making and operational efficiency.

It provides a centralized view of business performance across multiple dimensions, allowing users to quickly identify trends and patterns. The interactive dashboards support dynamic filtering and drill-down analysis for deeper insights into data. Overall, the solution enhances data accessibility and enables faster, more informed strategic decisions.

Problem Statement

- Lack of centralized reporting system
- Difficulty in tracking sales performance across categories
- Inefficient inventory monitoring
- Limited visibility into customer behavior
- Manual reporting leading to delays and errors

Objectives

- To build an interactive business intelligence dashboard
- To analyze sales, customer, and inventory data
- To track KPIs such as revenue, profit, and orders
- To identify top-performing products and customers
- To improve data-driven decision-making

Data Source

Excel was used as the data source with multiple tables:

- Sales Table → Orders, quantity, customer, product, date
- Products Table → Product details and category
- Customers Table → Customer information
- Inventory Table → Stock availability
- Regions Table → Country and region mapping
- Targets Table → Monthly revenue goals

Data Model

A star schema structure was used:

- Fact Table: Sales
- Dimension Tables: Products, Customers, Inventory, Regions

Relationships were created using:

- ProductID
- CustomerID
- Country

Key Metrics (DAX)

- Total Revenue = Quantity × Price
- Total Orders = Count of Order IDs
- Total Customers = Unique customers
- Total Profit = Revenue × 25%
- Profit Margin % = Profit ÷ Revenue
- Revenue Target = Sum of target values
- Achievement % = Revenue ÷ Target

Dashboard Pages

Executive Overview

This page provides a high-level summary of business performance using KPIs and trends.

Visuals:

- Total Revenue, Profit, Orders, Customers
- Revenue Trend (Line Chart)
- Sales by Category (Donut Chart)
- Sales by Region (Bar Chart)
- Revenue Achievement Gauge

Sales Performance

Focuses on product and sales analysis.

Visuals:

- Top 10 Products
- Top 10 Customers
- Salesperson Performance
- Category Revenue Breakdown
- Monthly Revenue Trend

Customer Insights

Analyzes customer behavior and segmentation.

Visuals:

- Revenue by Country (Map)
- Top Customers
- Customer Distribution
- Regional Performance
- Customer Performance Table

Inventory & Operations

Monitors stock levels and inventory health.

Visuals:

- Total Stock KPI

- Stock by Category
- Inventory Utilization Gauge
- Product Inventory Table
- Revenue vs Stock Analysis

Key Insights

- Electronics category contributes the highest revenue share
- Top 10 customers generate a significant portion of total revenue
- Inventory levels are balanced across major categories
- Certain regions outperform others in sales contribution
- Seasonal trends impact monthly revenue fluctuations
- 10. Tools & Technologies
- Microsoft Excel (Data Preparation)
- Microsoft Power BI (Visualization)
- DAX (Data Analysis Expressions)
- Data Modeling (Star Schema Design)

Outcome

The RetailX Analytics Suite successfully transforms raw sales data into meaningful business insights. The dashboard enables stakeholders to monitor performance, track KPIs, and make data-driven decisions efficiently.

It consolidates multiple data sources into a single unified reporting system for better visibility. The use of interactive visuals helps simplify complex datasets into clear and actionable insights. This ensures faster understanding of business performance and supports continuous improvement in operations.

Conclusion

This project demonstrates an end-to-end business intelligence workflow, from raw data preparation to interactive dashboard creation

in Power BI. The RetailX Analytics Suite integrates multiple Excel datasets into a structured star schema model, enabling efficient analysis and reporting. Key business KPIs such as revenue, profit, orders, customers, and target achievement are calculated using DAX to provide dynamic insights.

The dashboard improves decision-making by converting complex data into simple, interactive visuals. It is organized into multiple pages for sales, customers, and inventory, reflecting real-world reporting structure. Overall, the project highlights practical skills in data modeling, DAX, and dashboard design for business insights.